SIMATS ENGINEERING
CO3-ASSESSMENT TOOL 1
Experiments

COURSE NAME: Computer Networks **COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Dr.V.Parthipan

Code of Conduct:

I M.HARINI(192424011) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student HARINI.M

Criteria	Understanding of Concepts 2.5	Experimental Design 3	Use of Tools 2	Analysis of Results 1.5	Documentation 1	Total (10)
Q1						
Q2						
Q3						
Q4						

Faculty Signature

Experiments

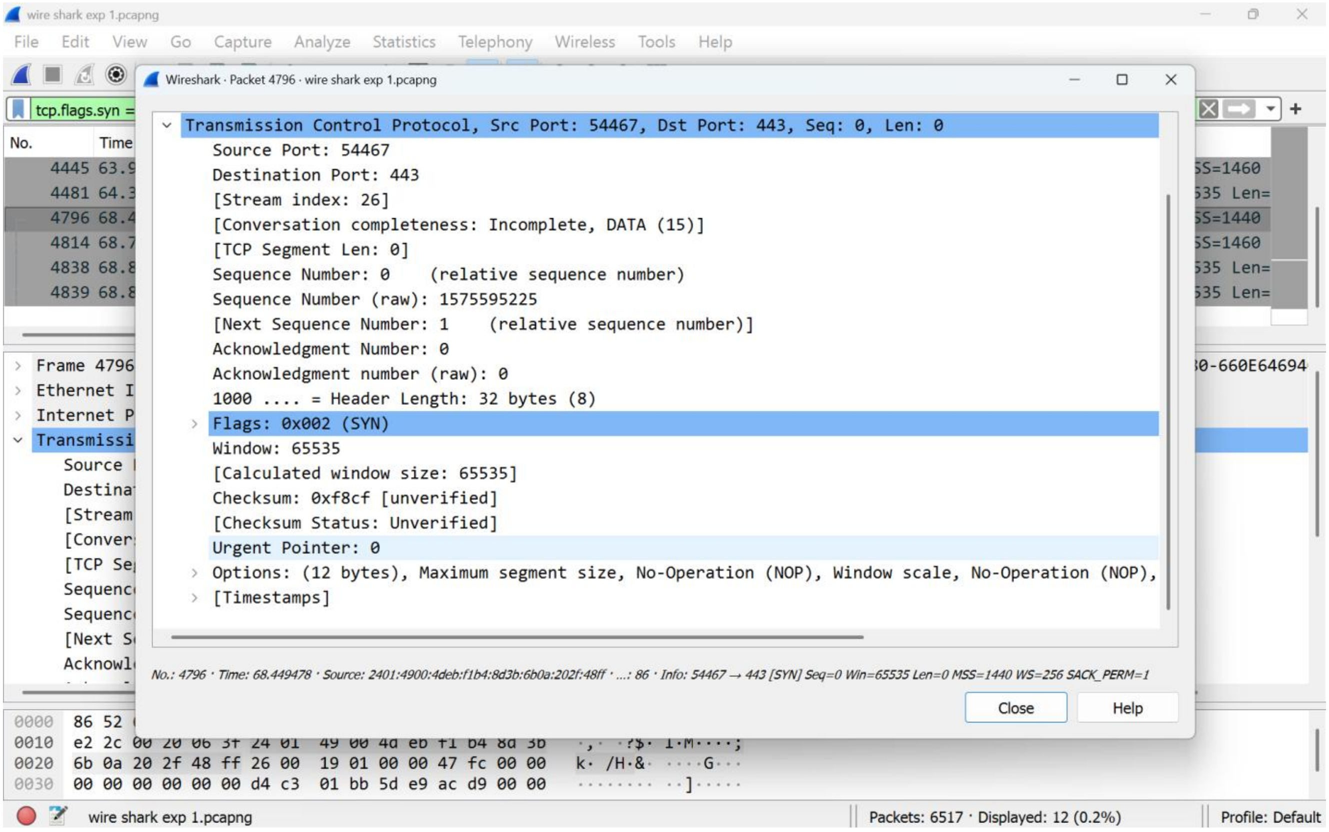
QUESTIONS:4 **Total Marks:40**

1. Capture packets during a TCP connection setup. Identify the three packets involved in the handshake (SYN, SYN-ACK, and ACK). Demonstrate the role of each flag and how they establish a reliable connection. (BL4) (10)
2. Capture a DNS query using UDP in Wireshark. Compare the UDP header size with TCP and note the absence of sequence numbers or acknowledgments. Discuss how this lightweight design impacts speed and reliability. (BL4) (10)
3. Simulate packet loss (e.g., disconnect briefly) and capture traffic in Wireshark. Observe how TCP retransmits lost packets while UDP does not. Explain why this difference makes TCP reliable but slower compared to UDP. (BL4) (10)
4. Use Wireshark to capture DNS traffic while resolving a domain name. Identify the query type (A, AAAA, MX) and the response received. Measure the time taken for resolution and explain why DNS typically uses UDP.(BL4) (10)

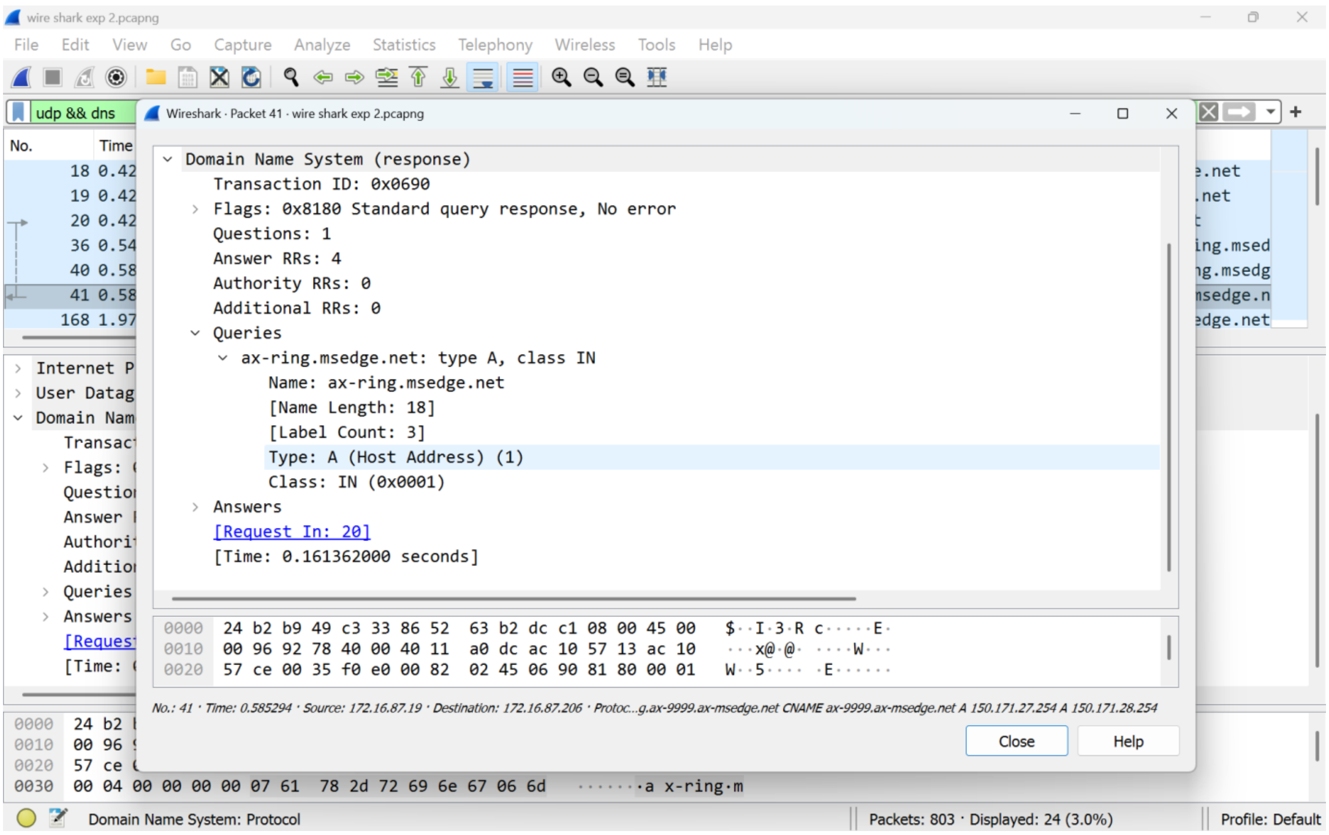
RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

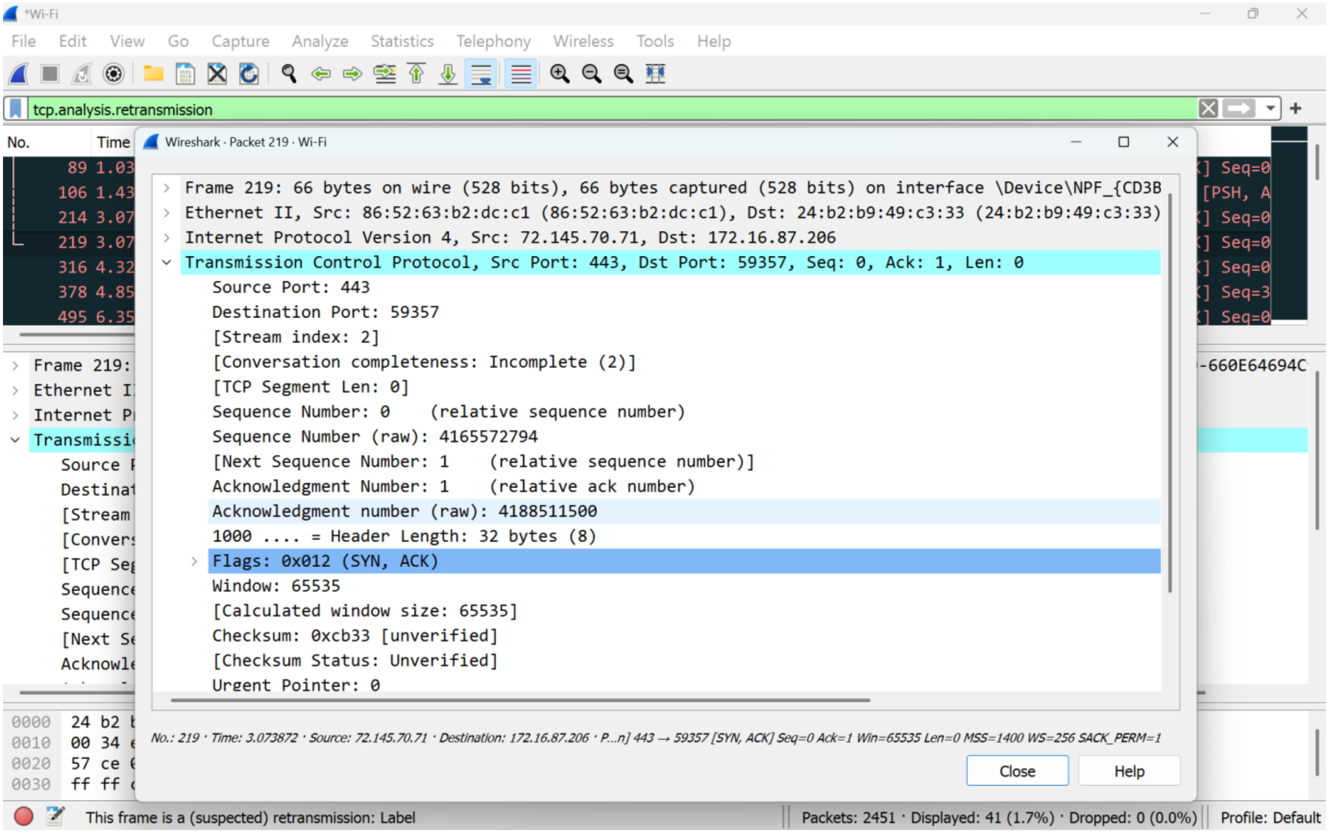
EXP 1


wire shark exp 1.pcapng
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.flags.syn
Wireshark · Packet 4796 · wire shark exp 1.pcapng
Transmission Control Protocol, Src Port: 54467, Dst Port: 443, Seq: 0, Len: 0
Source Port: 54467
Destination Port: 443
[Stream index: 26]
[Conversation completeness: Incomplete, DATA (15)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 1575595225
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 0
Acknowledgment number (raw): 0
1000 .... = Header Length: 32 bytes (8)
Flags: 0x002 (SYN)
Window: 65535
[Calculated window size: 65535]
Checksum: 0xf8cf [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), Maximum segment size, No-Operation (NOP), Window scale, No-Operation (NOP),
[Timestamps]
No.: 4796 · Time: 68.449478 · Source: 2401:4900:4deb:f1b4:8d3b:6b0a:202f:48ff · ...: 86 · Info: 54467 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM=1
Close
Help
wire shark exp 1.pcapng
Packets: 6517 · Displayed: 12 (0.2%)
Profile: Default

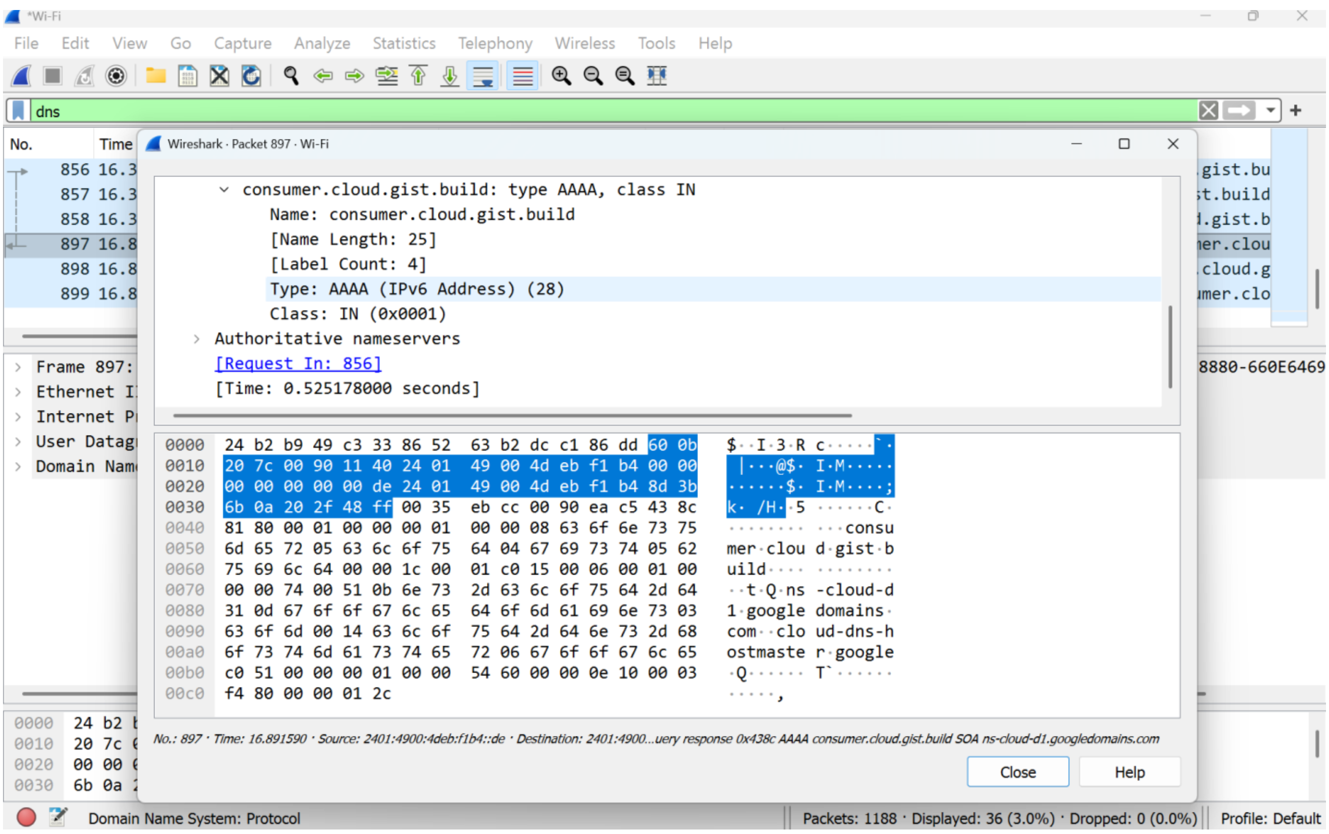
EXP 2


wire shark exp 2.pcapng
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
udp && dns
Wireshark · Packet 41 · wire shark exp 2.pcapng
Domain Name System (response)
Transaction ID: 0x0690
Flags: 0x8180 Standard query response, No error
Questions: 1
Answer RRs: 4
Authority RRs: 0
Additional RRs: 0
Queries
ax-ring.msedge.net: type A, class IN
Name: ax-ring.msedge.net
[Name Length: 18]
[Label Count: 3]
Type: A (Host Address) (1)
Class: IN (0x0001)
Answers
[Request In: 20]
[Time: 0.161362000 seconds]
0000 24 b2 b9 49 c3 33 86 52 63 b2 dc c1 08 00 45 00
0010 00 96 92 78 40 00 40 11 a0 dc ac 10 57 13 ac 10
0020 57 ce 00 35 f0 e0 00 82 02 45 06 90 81 80 00 01
No.: 41 · Time: 0.585294 · Source: 172.16.87.19 · Destination: 172.16.87.206 · Protoc...g.ax-9999.ax-msedge.net CNAME ax-9999.ax-msedge.net A 150.171.27.254 A 150.171.28.254
Close
Help
Domain Name System: Protocol
Packets: 803 · Displayed: 24 (3.0%)
Profile: Default

EXP 3


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.analysis.retransmission
Wireshark · Packet 219 · Wi-Fi
Frame 219: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{CD3B
Ethernet II, Src: 86:52:63:b2:dc:c1 (86:52:63:b2:dc:c1), Dst: 24:b2:b9:49:c3:33 (24:b2:b9:49:c3:33)
Internet Protocol Version 4, Src: 72.145.70.71, Dst: 172.16.87.206
Transmission Control Protocol, Src Port: 443, Dst Port: 59357, Seq: 0, Ack: 1, Len: 0
Source Port: 443
Destination Port: 59357
[Stream index: 2]
[Conversation completeness: Incomplete (2)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 4165572794
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 4188511500
1000 .... = Header Length: 32 bytes (8)
Flags: 0x012 (SYN, ACK)
Window: 65535
[Calculated window size: 65535]
Checksum: 0xcb33 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
No.: 219 · Time: 3.073872 · Source: 72.145.70.71 · Destination: 172.16.87.206 · P...n] 443 → 59357 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1400 WS=256 SACK_PERM=1
Close
Help
This frame is a (suspected) retransmission: Label
Packets: 2451 · Displayed: 41 (1.7%) · Dropped: 0 (0.0%)
Profile: Default

EXP 4


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dns
Wireshark · Packet 897 · Wi-Fi
consumer.cloud.gist.build: type AAAA, class IN
Name: consumer.cloud.gist.build
[Name Length: 25]
[Label Count: 4]
Type: AAAA (IPv6 Address) (28)
Class: IN (0x0001)
Authoritative nameservers
[Request In: 856]
[Time: 0.525178000 seconds]
0000 24 b2 b9 49 c3 33 86 52 63 b2 dc c1 86 dd 60 0b
0010 20 7c 00 90 11 40 24 01 49 00 4d eb f1 b4 00 00
0020 00 00 00 00 00 de 24 01 49 00 4d eb f1 b4 8d 3b
0030 6b 0a 20 2f 48 ff 00 35 eb cc 00 90 ea c5 43 8c
0040 81 80 00 01 00 00 00 01 00 00 08 63 6f 6e 73 75
0050 6d 65 72 05 63 6c 6f 75 64 04 67 69 73 74 05 62
0060 75 69 6c 64 00 00 1c 00 01 c0 15 00 06 00 01 00
0070 00 00 74 00 51 0b 6e 73 2d 63 6c 6f 75 64 2d 64
0080 31 0d 67 6f 6f 67 6c 65 64 6f 6d 61 69 6e 73 03
0090 63 6f 6d 00 14 63 6c 6f 75 64 2d 64 6e 73 2d 68
00a0 6f 73 74 6d 61 73 74 65 72 06 67 6f 6f 67 6c 65
00b0 c0 51 00 00 00 01 00 00 54 60 00 00 0e 10 00 03
00c0 f4 80 00 00 01 2c
No.: 897 · Time: 16.891590 · Source: 2401:4900:4deb:f1b4::de · Destination: 2401:4900...uery response 0x438c AAAA consumer.cloud.gist.build SOA ns-cloud-d1.googledomains.com
Close
Help
Domain Name System: Protocol
Packets: 1188 · Displayed: 36 (3.0%) · Dropped: 0 (0.0%)
Profile: Default